

1. Decomposition
2. $\text{N}_2 + 4\text{O}_2 + 3\text{H}_2 \rightarrow 2\text{HNO}_3 + 2\text{H}_2\text{O}$
3. $3.5 \times 10^{-3} \text{ g}$
4. 19.39 g
5. 12.9 g
6. 84%
7. molecular, acid
8. $\text{Mg}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Mg}(\text{OH})_2(\text{s})$
9. $3.62 \times 10^{-13} \text{ moles}$
10. Oxidized: Al Reduced: Cu
11. $\text{S}_8 + 12\text{O}_2 + 8\text{H}_2\text{O} \rightarrow 8\text{H}_2\text{SO}_4$
12. Arsenic
13. 8.79 g
14. $1.06 \times 10^0 \text{ mol}$
15. $\text{Ca}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2(\text{g}) + \text{Ca}(\text{OH})_2(\text{aq})$
16. $9.1 \times 10^{-50} \text{ moles}$
17. SCl_2
18. A_2B_3
19. 0.552 M
20. $\text{S}_8 + 12\text{O}_2 + 8\text{H}_2\text{O} \rightarrow 8\text{H}_2\text{SO}_4$
21. 279.9 mL
22. $\text{HBr} + \text{NaOH} \rightarrow \text{H}_2\text{O} + \text{NaBr}$
23. 6.157 g
24. No reaction occurs.
25. Oxygen
26. $2\text{ZnS} + 2\text{C} + 3\text{O}_2 \rightarrow 2\text{CO} + 2\text{SO}_2 + 2\text{Zn}$
27. $9.9 \frac{\text{mol}}{\text{L}}$
28. $\text{C}_3\text{H}_8 + 6\text{H}_2\text{O} \rightarrow 10\text{H}_2 + 3\text{CO}_2$
29. 44.0 mL
30. $5\text{O}_2 + \text{P}_4 + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$
31. 0.88 mL of ammonium chromate
32. $1.077 \times 10^{35} \text{ g}$
33. $3.28 \times 10^0 \text{ mol}$
34. $\text{C}_3\text{H}_8 + 6\text{H}_2\text{O} \rightarrow 10\text{H}_2 + 3\text{CO}_2$
35. $6\text{MnCO}_3 + 8\text{Al} + 3\text{O}_2 \rightarrow 4\text{Al}_2\text{O}_3 + 6\text{CO}_2 + 6\text{Mn}$
36. oxidized
37. $3\text{Sr}^{2+}(\text{aq}) + 2\text{PO}_4^{3-}(\text{aq}) \rightarrow \text{Sr}_3(\text{PO}_4)_2(\text{s})$
38. 5.04 g
39. 26.9%
40. C_6H_{14}
41. 0.280 moles
42. $2.53 \times 10^{51} \text{ g}$
43. Oxidized: Na Reduced: Cu

44. 0.382 M
45. 1.21 g
46. 9.5 mL
47. 27.19 mL
48. $5\text{O}_2 + \text{P}_4 + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$
49. 12.93 g
50. 7.411 mol
51. 616 kg
52. 0.0653 M
53. molecular
54. $\text{H}_2\text{SeO}_4 + \text{Ca}(\text{OH})_2 \rightarrow \text{CaSeO}_4 + 2\text{H}_2\text{O}$
55. 10.10 g
56. 3.944×10^{-3} moles
57. 271 mL
58. 1.0×10^{-57} moles
59. $3\text{C} + 4\text{Al}(\text{OH})_3 \rightarrow 3\text{CO}_2 + 4\text{Al} + 6\text{H}_2\text{O}$
60. Oxidized: Sr Reduced: Pb
61. 1.384×10^{-14} g
62. 0.11 mol
63. $\text{Ni}(\text{OH})_2 + 2\text{HClO} \rightarrow \text{Ni}(\text{ClO})_2 + 2\text{H}_2\text{O}$
64. 9.9 mM
65. 1.1×10^{-4} g
66. $2\text{C} + 2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 4\text{Cu} + 2\text{SO}_2 + 2\text{CO}$
67. 46.26%
68. 1.88×10^2 g
69. 47.0 g/mol
70. 14.40 g
71. $0.195 \frac{\text{mol}}{\text{L}}$
72. 2.0×10^{26} g
73. 0.2665 moles
74. BN
75. 0.0296 moles
76. 0.1866 g
77. 77%
78. $\text{C}_3\text{H}_8 + 6\text{H}_2\text{O} \rightarrow 10\text{H}_2 + 3\text{CO}_2$
79. 17%
80. $\text{Cu}^{2+}(\text{aq}) + \text{O}^{2-}(\text{aq}) \rightarrow \text{CuO}(\text{s})$
81. 0.26 mL of magnesium iodide
82. 1.1 M
83. 0.030 g
84. Oxidizing Agent: CO_2 Reducing Agent: K
85. 11.3 g
86. $6\text{MnCO}_3 + 8\text{Al} + 3\text{O}_2 \rightarrow 4\text{Al}_2\text{O}_3 + 6\text{CO}_2 + 6\text{Mn}$
87. H_3OF

88. 203.5 g
89. $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
90. $1.36 \times 10^2 \text{ g}$
91. No reaction occurs.
92. $\text{C}_9\text{H}_{18}\text{O}$
93. $\text{H}_3\text{AsO}_4 + \text{Bi}(\text{OH})_3 \rightarrow \text{BiAsO}_4 + 3\text{H}_2\text{O}$
94. $x = 4$
95. $5\text{O}_2 + \text{P}_4 + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$
96. $\text{N}_2 + 4\text{O}_2 + 3\text{H}_2 \rightarrow 2\text{HNO}_3 + 2\text{H}_2\text{O}$
97. $2.62 \times 10^2 \text{ g}$
98. 4.6767 mol
99. $0.022 \frac{\text{mol}}{\text{L}}$
100. $\text{Sr}(s) + 2\text{H}_2\text{O}(l) \rightarrow \text{Sr}(\text{OH})_2(aq) + \text{H}_2(g)$
101. 1.2 mL
102. Fe_2S_3 and H_2O
103. $\text{CH}_3\text{OH} + \text{CO} \rightarrow \text{CH}_3\text{COOH}$
104. 32%
105. 874 kg
106. $\text{TiO}_2 + 2\text{Cl}_2 + \text{C} + 2\text{Mg} \rightarrow \text{Ti} + \text{CO}_2 + 2\text{MgCl}_2$
107. 728 kg
108. 98.0 g/mol
109. $\text{Mg}_3(\text{PO}_4)_2$ and H_2O
110. $2\text{C}_9\text{H}_{19}\text{OH} + 27\text{O}_2 \rightarrow 20\text{H}_2\text{O} + 18\text{CO}_2$
111. 6.094×10^{21} oxygen atoms
112. $3.08 \times 10^1 \text{ g}$
113. 90%
114. 47.56 g
115. $\text{CH}_3\text{OH} + \text{CO} \rightarrow \text{CH}_3\text{COOH}$
116. 2.2968 mol
117. 0.0156 mol
118. 1.6×10^{-2} moles
119. 12
120. $\text{HCl}(aq) + \text{H}_2\text{O}(\ell) \rightarrow \text{Cl}^-(aq) + \text{H}_3\text{O}^+(aq)$
121. 2.88×10^{20} germanium atoms
122. 15%
123. Fe_2S_3 and H_2O
124. $\text{Sr}(s) + 2\text{H}_2\text{O}(l) \rightarrow \text{Sr}(\text{OH})_2(aq) + \text{H}_2(g)$
125. $1.86 \times 10^1 \text{ mol}$
126. $\text{Al}_2\text{S}_3 + 6\text{H}_3\text{OI} \rightarrow 3(\text{H}_3\text{O})_2\text{S} + 2\text{AlI}_3$
127. $2\text{HNO}_2 + \text{Ca}(\text{OH})_2 \rightarrow \text{Ca}(\text{NO}_2)_2 + 2\text{H}_2\text{O}$
128. 0 M
129. $3\text{C} + 4\text{Al}(\text{OH})_3 \rightarrow 3\text{CO}_2 + 4\text{Al} + 6\text{H}_2\text{O}$
130. 21.6 g
131. Double-Displacement
132. $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$

133. 4.0×10^{23} boron atoms
134. $161.4 \frac{\text{g}}{\text{mol}}$
135. Oxidizing Agent: H_2O Reducing Agent: Ba
136. 7.08 g
137. $\text{CH}_3\text{OH} + \text{CO} \rightarrow \text{CH}_3\text{COOH}$
138. ZnSO_3
139. $2\text{ZnS} + 2\text{C} + 3\text{O}_2 \rightarrow 2\text{CO} + 2\text{SO}_2 + 2\text{Zn}$
140. Bromine
141. $2\text{KOH} + \text{H}_2\text{SO}_4 \rightarrow \text{K}_2\text{SO}_4 + 2\text{H}_2\text{O}$
142. 30.6 g
143. 0.1074 moles
144. 0.259 mol
145. molecular, acid
146. 0.0878 moles
147. $6\text{MnCO}_3 + 8\text{Al} + 3\text{O}_2 \rightarrow 4\text{Al}_2\text{O}_3 + 6\text{CO}_2 + 6\text{Mn}$
148. 0.31 g
149. 60.2 g
150. 69%
151. 0.6872 moles
152. $\text{N}_2 + 4\text{O}_2 + 3\text{H}_2 \rightarrow 2\text{HNO}_3 + 2\text{H}_2\text{O}$
153. $0.115 \frac{\text{mol}}{\text{L}}$
154. 16.01 mL
155. 0.021 g
156. -2
157. molecular, acid
158. 63%
159. 1.37×10^{-24} moles
160. 22
161. 1.02×10^1 g
162. 82.0%
163. 1.43×10^{-17} moles
164. 10
165. $\text{TiO}_2 + 2\text{Cl}_2 + \text{C} + 2\text{Mg} \rightarrow \text{Ti} + \text{CO}_2 + 2\text{MgCl}_2$
166. 7.66 mL
167. 0.0475 L
168. KF
169. 1.000 moles
170. 1.9×10^9 g
171. 0.689 M
172. $3.80 \times 10^{-4} \frac{\text{mol}}{\text{L}}$
173. 63.0 g/mol
174. 0.04317 moles
175. 13.2 g
176. $x = 3$
177. 0.542 moles

178. $704.9 \frac{\text{g}}{\text{mol}}$
179. $3\text{HClO} + \text{Bi}(\text{OH})_3 \rightarrow \text{Bi}(\text{ClO})_3 + 3\text{H}_2\text{O}$
180. $88.005 \frac{\text{g}}{\text{mol}}$
181. Double-Displacement
182. 8.169 g
183. 1.2%
184. $5.47 \times 10^{10} \text{ g}$
185. 5.34 g
186. $3\text{Ca}(\text{OH})_2 + 2\text{H}_3\text{BO}_3 \rightarrow \text{Ca}_3(\text{BO}_3)_2 + 6\text{H}_2\text{O}$
187. 17.6 mM
188. $4.624 \times 10^{50} \text{ g}$
189. $\text{Cu}(\text{OH})_2 + 2\text{HI} \rightarrow 2\text{H}_2\text{O} + \text{CuI}_2$
190. $2\text{C} + 2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 4\text{Cu} + 2\text{SO}_2 + 2\text{CO}$
191. -2
192. $2\text{Bi}(s) + 3\text{Pt}^{2+}(aq) \rightarrow 2\text{Bi}^{3+}(aq) + 3\text{Pt}(s)$
193. 0.125 mL of ammonium chromate
194. $2\text{ZnS} + 2\text{C} + 3\text{O}_2 \rightarrow 2\text{CO} + 2\text{SO}_2 + 2\text{Zn}$
195. Oxidizing Agent: AuOH Reducing Agent: Fe
196. oxidized
197. $2\text{C} + 2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 4\text{Cu} + 2\text{SO}_2 + 2\text{CO}$
198. 35.1%
199. $\text{HNO}_3(aq) + \text{H}_2\text{O}(\ell) \rightarrow \text{NO}_3^-(aq) + \text{H}_3\text{O}^+(aq)$
200. 120. g